

O RESONATOR

Professional User Guide, Tutorial & FAQ

For the Code O rotating string instrument and effect

IUS Music

Research prototype documentation

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1. What O Resonator is

O Resonator (also called **Code O**) is a minimalist electronic rotating string instrument and audio effect. Its visual and sonic model is a black circular disk containing seven internal concentric string rings.

Each string ring has an electronic input path. When a ring is active, its input path opens and the string contributes to the rotating resonant body. When a ring is inactive, its path is closed and the string is silent or nearly silent.

The flywheel does not pluck, strike or trigger the strings. It continuously modulates the active strings through motion, pressure, inertia, damping, brightness and body resonance. This creates sustained, blooming and evolving tones rather than note sequences.

What it is not

- Not a plucked string instrument.
- Not an arpeggiator or step sequencer.
- Not a clock-hand trigger wheel.
- Not a bridge, grip, latch or contact-point mechanism.
- Not a synth that generates random notes from rotation alone.

The essential rule is simple: **rotation shapes active strings; it does not create notes by itself.**

Operating modes

Mode	Use
Standalone	Internal sustained instrument
Effect plug-in	Processes external audio

In standalone mode, active strings are excited internally. In effect mode, external audio is routed through the active resonant paths and then shaped by the same flywheel and body system.

2. Core idea: strings, flywheel and body

Seven string rings

The instrument contains seven internal string rings. In engine order, they are:

A1, D2, G2, C3, F3, A3, D4

In the side-view visualiser and resonator bank, they are shown from outside to inside:

D4, A3, F3, C3, G2, D2, A1

The strings are concentric rings, not radial spokes. Active rings contribute to the same resonant body. They do not play one after another unless you deliberately automate activation or note assignment.

Activation model

Activating a string opens that string's input path smoothly. The string becomes part of the body resonance. Deactivating a string closes the path and the string fades or damps away while the other active strings continue.

Changing active strings should feel like a tonal arrival or removal, not a pluck. The design goal is continuous resonance, not transient attack.

Standalone signal flow

1. Active string input
2. Internal sustained excitation
3. Tuned string resonance
4. Flywheel modulation
5. Circular body resonance
6. Master output and limiter

If no strings are active, standalone mode should be silent or nearly silent. Rotation alone should not create a pitched note.

Effect signal flow

7. Host audio input
8. Input gain and isolation
9. Seven string sends
10. Active string gates
11. Tuned resonant paths
12. Flywheel modulation
13. Body resonance
14. Mix, master and limiter

In effect mode, no external input means there is no main processed sound unless Internal Sustain is explicitly enabled. Input plus no active strings should be silent or nearly silent, because all string paths are closed.

3. Quick start tutorials

A. Create a dark drone

15. Activate D2 and A3.
16. Turn Disk On.
17. Set Sustain Feed to a medium value.
18. Raise Pressure until the tone blooms.
19. Set Flywheel Depth to a medium value.
20. Increase Flywheel Inertia for slow, heavy motion.
21. Add Body Move or Character Amount until the sound feels full.
22. Set Master to a safe output level.

Expected sound: a continuous dark drone that blooms into motion. It should not tick, pluck or play a sequence.

B. Create evolving harmony

23. Activate three rings such as D2, F3 and A3.
24. Choose an Evolution mode.
25. Increase Mod Depth gradually.
26. Set Period to several rotations.

27. Use Flywheel Inertia to smooth the movement.

28. Keep Texture and Body Grit low for a clean harmonic body.

Expected sound: the active rings merge into one moving body. The harmony should evolve as a single resonance, not as a note order.

C. Process external audio

29. Insert O Resonator as an effect on an audio track.

30. Send audio into the plug-in.

31. Activate one or more string rings.

32. Raise Input Gain.

33. Increase Input Isolation if you want stronger transformation.

34. Adjust Pressure, Damping and Flywheel Depth.

35. Use Mix to blend dry and processed signal.

Expected result: voice, guitar, synth, drums or field recordings are re-voiced through the selected circular string paths.

4. Reading the interface

Disk visualiser

The disk visualiser is a side view of the internal ring body. Dim rings are inactive. Bright red or blue rings indicate active strings. A moving energy mark shows flywheel motion on active rings. The mark is not a pluck point and does not trigger notes.

Resonator bank table

The resonator bank table shows each ring as a live row. It is not decorative. The table updates as strings are activated, notes are assigned, and the engine produces energy.

Column	Meaning
Ring	Visual index
Note	Current note
Freq	Current frequency
Level/Energy	Live output
Damp	Damping amount
State	Open or closed

If a ring is active, its state should read OPEN. If it is inactive, it should read CLOSED. Assigned MIDI notes should update the Note and Freq fields for the relevant ring.

Waveform monitors

Input Wave shows incoming audio. Output Wave shows the processed signal leaving the engine. In effect mode, these monitors are the fastest way to confirm that audio is entering and leaving the plug-in.

Modulation cards

The modulation cards are live parameter readouts. Mod Depth, Motion Drift, Body Move, Bright Move and Pressure reflect the current parameter values and their modulation curves.

5. Control reference

This section explains every main visible function. Use the controls musically: set the active strings first, then shape sustain, motion, body and output.

Disk and transport

Disk On - Starts or stops flywheel motion. When off, flywheel energy decays and the visual phase stops or freezes. When on, shared phase movement resumes.

Host Sync - Locks rotation to the host tempo when available. Disable it for free-running speed.

Speed - Sets the rotation rate when free-running. Slow values feel darker and steadier; faster values add energy, motion and brightness.

Time Mapping - Chooses how musical time maps to disk motion, such as bar or beat divisions.

Reset - Sets how phase reset behaves in relation to musical time. It should reset motion, not trigger a note.

Phase - Offsets the visual and modulation phase around the circular body. It changes position, not pitch.

Direction - Reverses the motion direction. It changes the rotation feel but does not reverse a melody or change the active notes.

Harmony and ring selection

Harmony - Selects a preset ring combination or manual control. Presets choose active resonant paths; they do not create sequences.

String activation buttons - Open or close the seven ring paths. Active strings contribute to sound; inactive strings stay silent or nearly silent.

Note assignment - Allows active strings to be retuned or fingered, usually by MIDI. Inactive strings should not sound from MIDI alone.

Motion and evolution

Rise - Adds rising rotational energy to the flywheel movement signal. Higher values increase motion intensity.

Flywheel - Controls how strongly rotation shapes the active string paths. Use it for circular pressure and movement.

Inertia - Controls how slowly the flywheel responds. Higher inertia creates a heavier, slower bloom.

Evolution - Chooses the type of long-term movement, such as rise, wave, freeze or random walk, depending on available modes.

Mod Depth - Sets the amount of evolution and modulation. Low values are stable; high values are more animated.

Period - Sets the duration of the evolution cycle in rotations or musical time.

Motion Drift - Adds rotation-linked timbre drift. Use it to make the tone move without sounding like a trigger.

String energy and response

Pressure - Main drive into the active strings. Higher values create stronger, brighter and more forceful resonance.

Sustain Feed - Continuous energy floor for active strings. Raise it when the sound is too weak or fades too quickly.

String Sustain - Controls how long string energy remains. Higher values ring longer.

Rub Energy - Adds continuous rub-like energy. Use gently for texture and sustain support.

Bend - Adds tension or deflection behaviour to active strings, especially when mapped to performance control.

Harmonics - Increases overtone emphasis. Use for shimmer, brightness or stronger harmonic presence.

Damping - Restrains ring time. Higher damping makes the sound shorter, tighter and darker.

Bright Move - Adds brightness movement linked to pressure or contact accent. Use low values for darker drones.

Input Response - Shapes how quickly input and activation respond. Use smoother response for less clicking and more natural blooms.

Body and texture

Body Move - Controls body resonance movement and density. Raise it to make the disk feel fuller and more physical.

Body Grit - Adds optional grit or saturation. Keep low for the pure rotating string sound.

Texture - Adds noise or friction colour. Use sparingly to avoid tick-like artifacts.

Width - Adds stereo spread or subtle phase movement. Use carefully so the body remains focused.

Quality - Adjusts processing quality. Higher settings may sound smoother but can use more CPU.

Oversampling - Reduces aliasing and improves high-energy processing at the cost of CPU and possible latency.

Input, mix and sidechain

Input Gain - Controls how much external audio enters the resonant string paths in effect mode.

Input Isolation - Controls how strongly incoming audio is forced through the active rings. Higher values transform the input more.

Mix - Blends processed output with the original input. Use fully wet for the pure O Resonator effect, or blend dry signal for clarity.

Sidechain Body - Feeds sidechain energy into the resonant body path when the host provides sidechain input.

Sidechain Pressure - Lets sidechain energy influence pressure and movement.

Internal Sustain - Adds optional internal sustain in effect mode. Leave off for a pure input-processing effect.

Space and output

Delay - Controls the wet amount of the optional delay. Delay is for space and rhythm, not the core sound identity.

Delay Time - Sets delay length when available.

Delay Feedback - Sets delay repeats when available.

Space - Controls the wet amount of the optional reverb. Add after the core body tone is balanced.

Space Size - Sets the reverb size when available.

Space Damping - Controls high-frequency decay in the reverb when available.

Master - Final output level. Adjust after the string, motion and mix levels are set.

Limiter - Final protection stage. Leave enabled when using high pressure, body or input settings.

6. Meters and live readouts

Use the live readouts to diagnose signal flow before increasing levels.

Readout	Use
Input meter	Incoming signal
Output meter	Final signal
Flywheel meter	Motion energy
Pressure card	Drive state
Ring state	Open/closed

If the Input Wave is moving but the Output Wave is not, check that at least one string is active and that Mix, Master and Input Gain are raised. If Output Wave moves but the sound is too quiet, raise Master gradually and check host output routing.

7. Recommended starting settings

Clean dark drone

Setting	Value
Active rings	D2 + A3
Speed	120 BPM
Flywheel	45%
Inertia	60%
Sustain Feed	55%
Pressure	45%
String Sustain	90%
Texture / Grit	0%
Space / Delay	0%

This is the safest starting point for the instrument's identity: dark, sustained, rotating and not plucked.

Evolving harmony

Setting	Value
Active rings	D2 + F3 + A3
Evolution	Slow
Mod Depth	35-60%
Period	Long
Body Move	Medium
Damping	Medium

External input processor

Setting	Value
Mode	Effect
Input Gain	Raise
Input Isolation	Medium-high
Internal Sustain	Off
Mix	Taste
Active rings	1-3 rings

8. FAQ

Why is there no sound?

Activate at least one ring. In effect mode, confirm audio is entering the plug-in, then raise Input Gain, Pressure, Sustain Feed and Master carefully.

Why does MIDI not play closed strings?

Closed strings have closed input paths. MIDI can retune or shape active strings, but it should not arm inactive rings by itself.

Is the moving mark a pluck point?

No. It is a motion/energy indicator. The flywheel shapes active strings continuously; it does not strike them.

Can I use it as a normal synth?

You can use standalone mode as an instrument, but it behaves like a sustained resonant body, not a note-triggered synth voice.

Can it process vocals, guitar or drums?

Yes. Use the effect version, activate rings, raise Input Gain and set Mix to taste.

Why does it sound too bright?

Lower Pressure, Bright Move, Harmonics, Texture and Body Grit. Increase Damping for a darker tone.

Why does it feel too static?

Increase Flywheel, Mod Depth, Motion Drift or Body Move. Choose a slower Evolution period for long movement.

Why does it feel too jumpy?

Increase Inertia and Input Response smoothing. Reduce Mod Depth and fast reset settings.

How do I keep it clean?

Use low Texture, low Body Grit, moderate Pressure, high Sustain, and add Space only after the core tone is balanced.

Are the table and visual meters real?

Yes. The table, polar graph, waveform monitors and modulation cards are live readouts connected to the processor state.

9. Rights and credits

O Resonator / Code O is a research and demonstration instrument created by Pezhman Farhangi under the IUS Music brand. The included software and documentation are provided for study, teaching, technical explanation, research demonstration and discussion.

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